

Problem 5418

Solution by: Simon Del Villar, UBC Okanagan, Kelowna, B.C.

Notice that the exponents are small: by AM–GM, $k^2 + n^2 \geq 2kn$, so

$$0 \leq \frac{k}{k^2 + n^2} \leq \frac{1}{2n}$$

for all $1 \leq k \leq n$. Since each exponent is at most $1/(2n) \rightarrow 0$, we can Taylor-expand $e^x = 1 + x + \frac{x^2}{2} + O(x^3)$ and substitute $x = k/(k^2 + n^2)$:

$$e^{k/(k^2+n^2)} = 1 + \frac{k}{k^2 + n^2} + \frac{1}{2} \left(\frac{k}{k^2 + n^2} \right)^2 + O\left(\frac{1}{n^3}\right).$$

Summing from $k = 1$ to n :

$$-n + \sum_{k=1}^n e^{k/(k^2+n^2)} = \sum_{k=1}^n \frac{k}{k^2 + n^2} + \frac{1}{2} \sum_{k=1}^n \left(\frac{k}{k^2 + n^2} \right)^2 + O\left(\frac{1}{n}\right).$$

The quadratic sum also vanishes as $n \rightarrow \infty$: since each term is at most $1/(2n)^2 = 1/(4n^2)$, summing n of them gives at most $1/(4n) \rightarrow 0$. So in the limit, only the linear sum survives:

$$\lim_{n \rightarrow \infty} \left(-n + \sum_{k=1}^n e^{k/(k^2+n^2)} \right) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k}{k^2 + n^2}.$$

To evaluate the right-hand side, factor n^2 from the denominator to recognize a Riemann sum:

$$\sum_{k=1}^n \frac{k}{k^2 + n^2} = \frac{1}{n} \sum_{k=1}^n \frac{k/n}{1 + (k/n)^2}.$$

This is exactly the Riemann sum for $\int_0^1 \frac{x}{1+x^2} dx$, so as $n \rightarrow \infty$ it converges to that integral. Using the antiderivative $\frac{1}{2} \ln(1+x^2)$:

$$\int_0^1 \frac{x}{1+x^2} dx = \frac{1}{2} \left[\ln(1+x^2) \right]_0^1 = \frac{\ln 2}{2}.$$

Therefore

$$\lim_{n \rightarrow \infty} \left(-n + \sum_{k=1}^n e^{k/(k^2+n^2)} \right) = \frac{\ln 2}{2}.$$

□